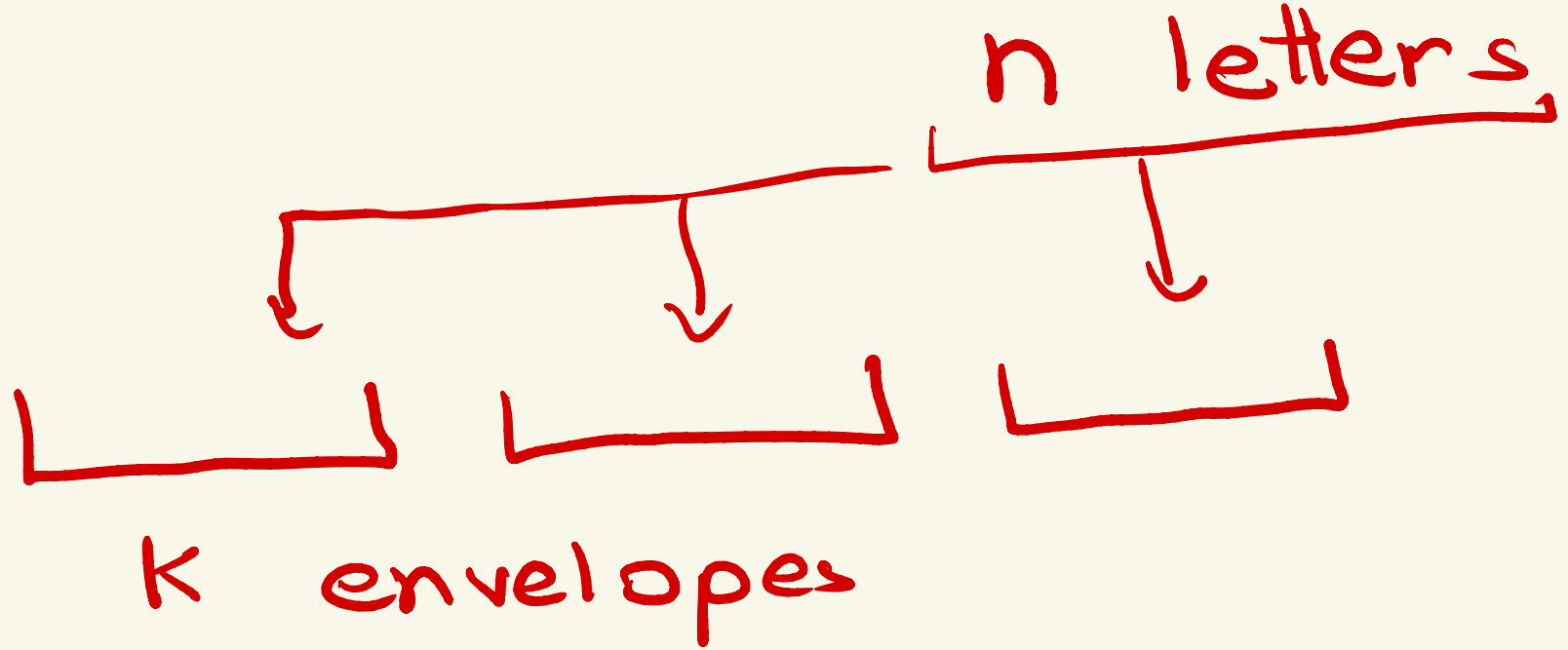


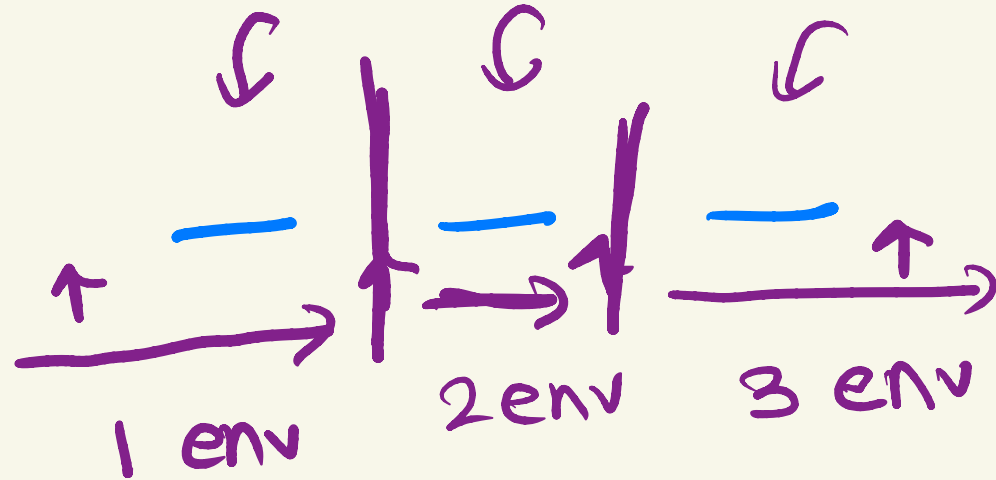
Combinatorics 2

Stars & Bars



3 letters

→ 3 envelopes



k envelopes

$k-1$ partition lines



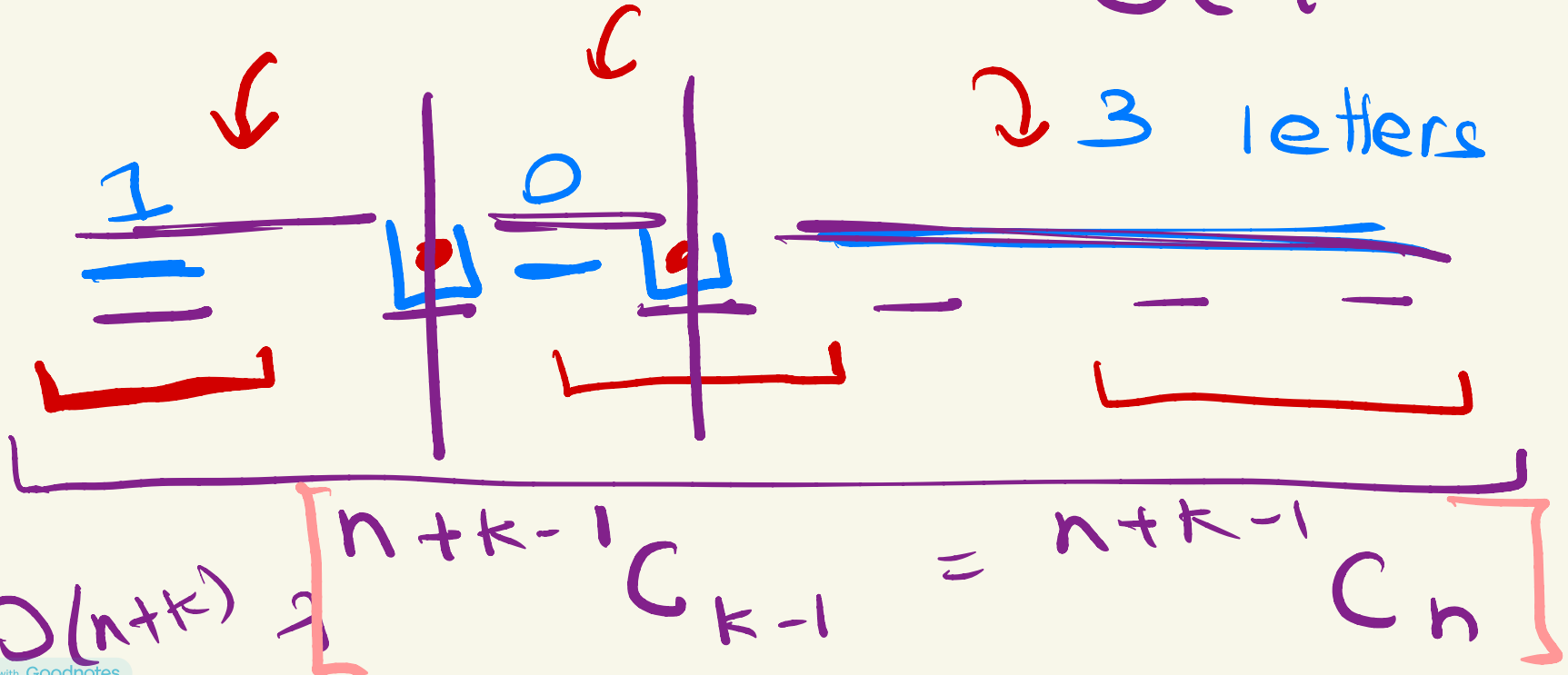
4 letters

3 envelopes

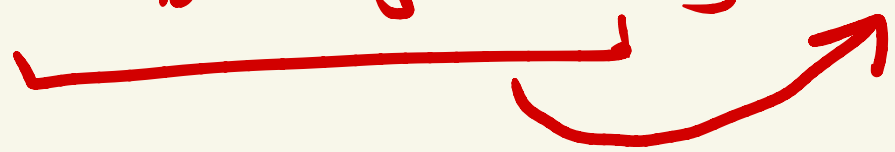
$n + k - 1$

$O(nk)$

3 letters



n ^{identical} objects, k containers



$$\underbrace{G_i \geq 0 \quad \sum C_i = n}$$

Stars and Bars

- Number of ways to n identical objects into k distinct boxes

$$\binom{n+k-1}{k-1} = \binom{n+k-1}{n}.$$

Stars and Bars Application

- What are the number of ways to solve the equation below?



$$\underline{x_1} + \underline{x_2} + \cdots + \underline{x_k} = n$$

identical
objects

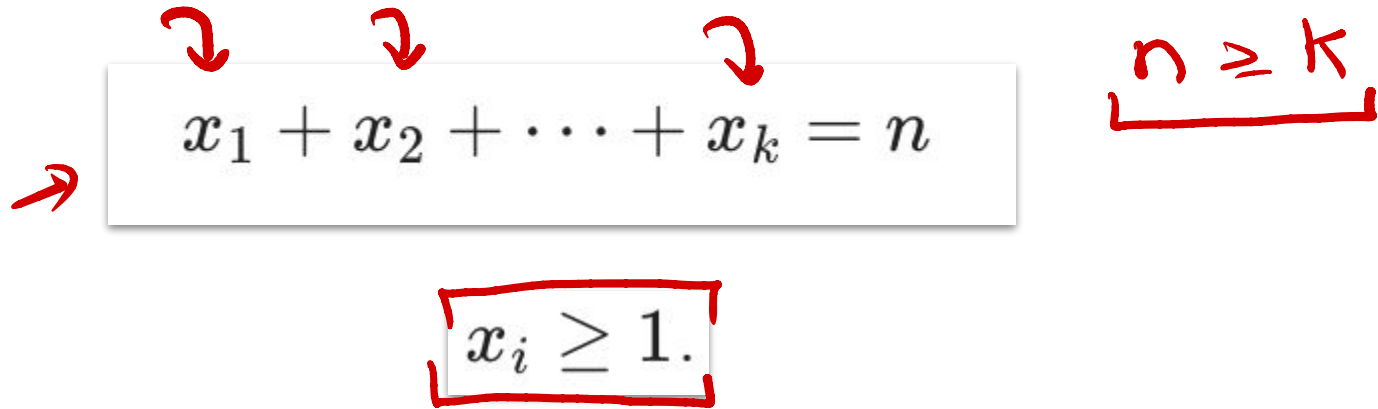


$$x_i \geq 0.$$

$$\binom{n+k-1}{k-1}$$

Stars and Bars Application

- What are the number of ways to solve the equation below?



The diagram illustrates the stars and bars problem. It features the equation $x_1 + x_2 + \cdots + x_k = n$ inside a light gray box. Four red arrows point to the variables x_1 , x_2 , x_k , and the entire equation. To the right of the equation box, the constraint $n \geq k$ is written in red and underlined. Below the equation box, the constraint $x_i \geq 1$ is written in black and enclosed in a red rectangular box.

$$x_1 + x_2 + \cdots + x_k = n$$

$n \geq k$

$x_i \geq 1$

$$x_i \geq 1$$

$$\begin{cases} x_1 = \underbrace{1} + \underbrace{x'_1} \geq 0 \\ x_2 = 1 + x'_2 \end{cases}$$

$$\left(\underset{\uparrow}{1} + x'_1 \right) + \left(\underset{\uparrow}{1} + x'_2 \right) + \dots + \left(\underset{\uparrow}{1} + x'_k \right) = n$$

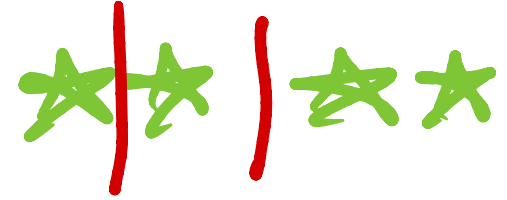
$$x'_i \geq 0$$

$$x'_1 + x'_2 + x'_3 \dots x'_k = n - k$$

$$x'_i \geq 0$$

$$\left(\begin{array}{c} \underbrace{(n-k)}_{\text{element}} + \underbrace{(k-1)}_{\text{partitions}} \\ \underbrace{k-1}_{\text{partitions}} \end{array} \right)$$

Stars and Bars Application



- What are the number of ways to solve the equation below?

$$x_1 + x_2 + \cdots + x_k = n$$

Handwritten annotations for the equation above:

- A large red arrow points to the equation.
- Red arrows point to the variables x_1 , x_2 , and x_k .
- Red brackets are drawn under x_1 and x_2 , with the numbers 4 and 7 written below them respectively.

Below the equation, a separate box contains the constraint:

$$x_i \geq a_i.$$

A red arrow points to the variable x_i in the constraint, and a red bracket is drawn under the entire constraint box.

$$x_1 + x_2 + x_3 = 5$$

$$x_1 \geq \underline{2} \leftarrow a_1$$

$$x_2 \geq \underline{0} \leftarrow a_2$$

$$x_3 \geq \underline{1} \leftarrow a_3$$

$$x_1 = (2 + x'_1)$$

$$x_2 = (0 + x'_2)$$

$$x_3 = (1 + x'_3)$$

$$x'_i \geq 0$$

$$\underset{\uparrow}{(a_1 + x'_1)} + \underset{\uparrow}{(a_2 + x'_2)} \dots \underset{\uparrow}{(a_k + x'_k)} = n$$

$$\underbrace{x'_1 + x'_2 \dots x'_k}_{x'_i \geq 0} = \underbrace{n - \sum a_i}$$

$$x'_i \geq 0$$

$$\binom{(n - \sum a_i) + (k-1)}{\underbrace{k-1}}$$



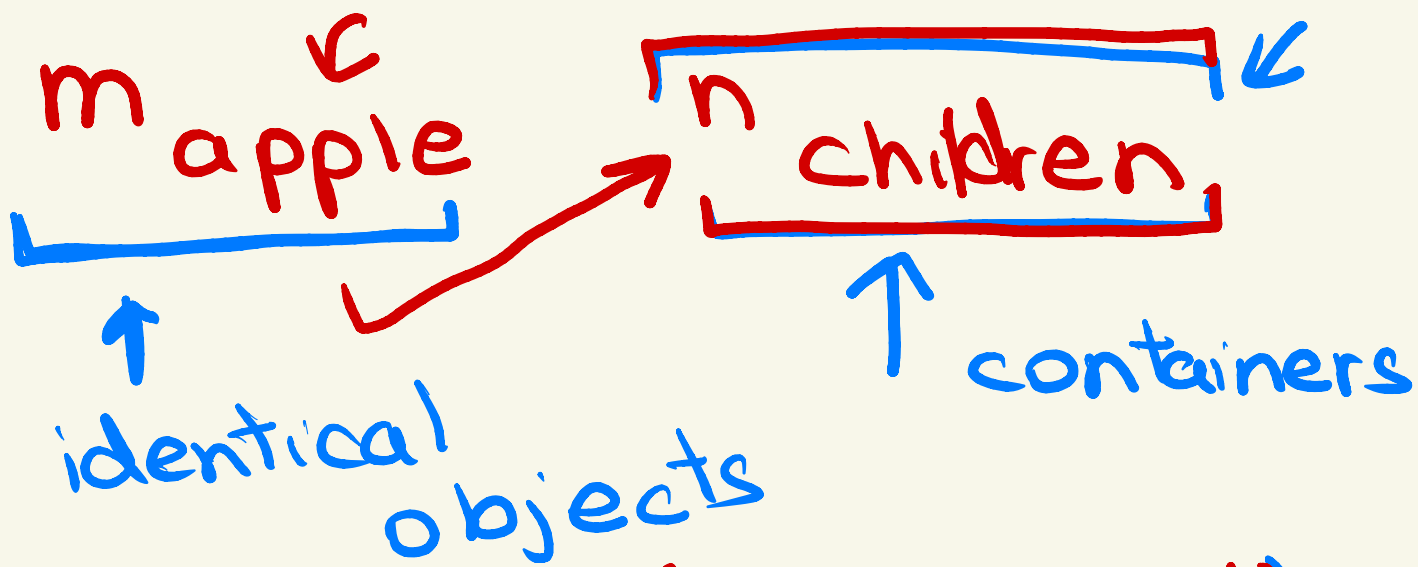
Let's solve some problems,



Distributing Apples - [Link](#)



Array - [Link](#)



$$\binom{m+n-1}{n-1}$$

$$2 \rightarrow \underline{4} \quad \% 10^a + 7$$

$$[1, 1] \quad [2, 2]$$

$$[2, 1] \quad [1, 2]$$

$$1 \leq x_i \leq n$$

— — — — — n

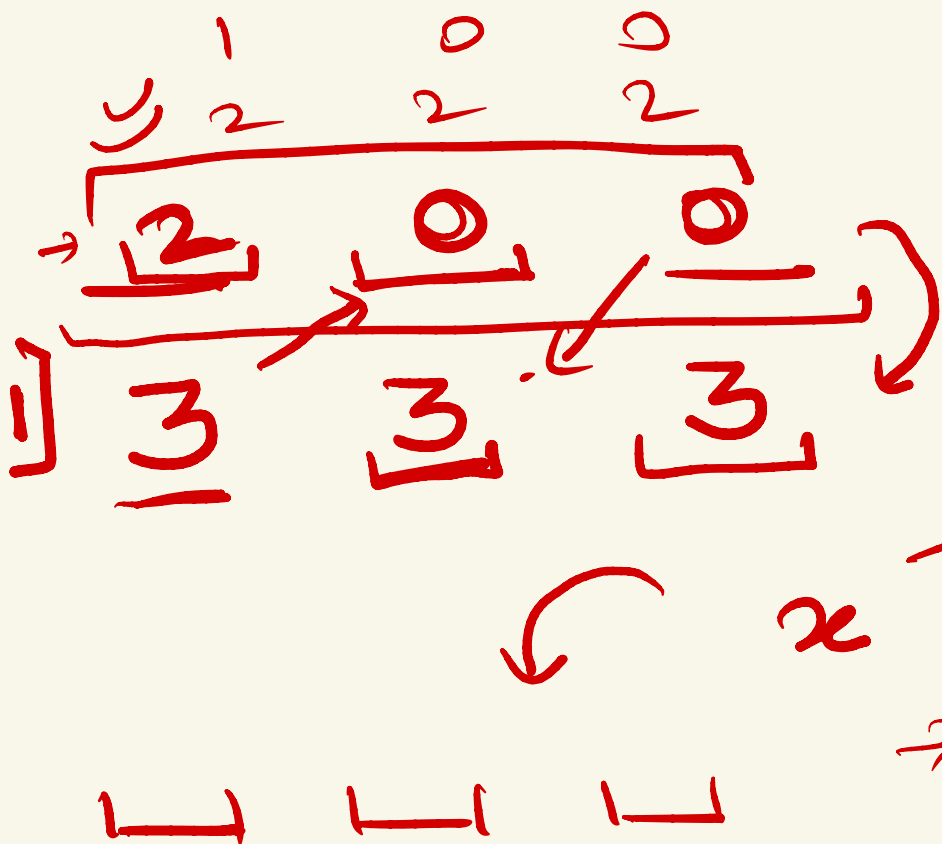
$$1 \leq x_i \leq n$$

↳ ascending order

↳ descending order

$$\begin{array}{ccccccc}
 \frac{1}{1} & \frac{2}{1} & \frac{2}{012} & \frac{4}{1} & \frac{5}{1} & \frac{6}{1} & \frac{7}{1} \\
 \underbrace{\quad} & & & & & & \\
 + & & & & & & \\
 \hookrightarrow 6 & & & & & & \\
 \boxed{17-1} & & & & & &
 \end{array}$$

|Ascending| = |Descending|

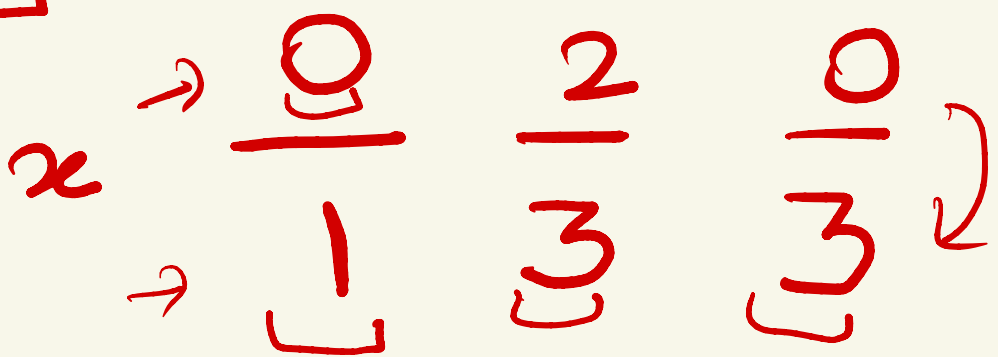


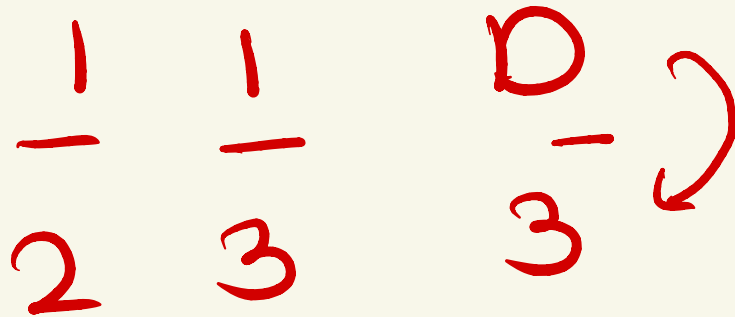
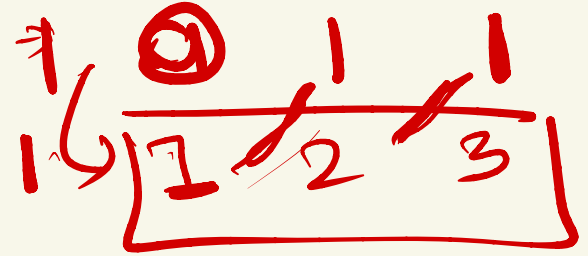
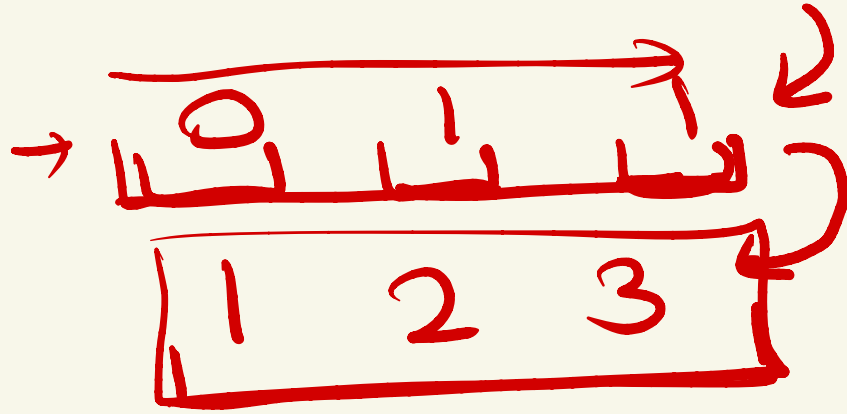
atmost $\lfloor n-1 \rfloor$ identical objects

$\downarrow$

$\boxed{2}$

n items





Distribute atmost $n-1$ objects
among n containers

$$\underline{0} > \underline{1} > \underline{2} \dots \underline{n-1} \quad \boxed{x}$$

$$\sum_{x=0}^{n-1} \left(\frac{x+n-1}{n-1} \right) * 2 - n \quad \% K$$

Problems on independent choices

- Lucky Numbers - [Link](#)
- Number of subsets of an array with at least 1 element
- Monotonic Remuneration - [Link](#)
- The Intriguing Obsession - [Link](#)

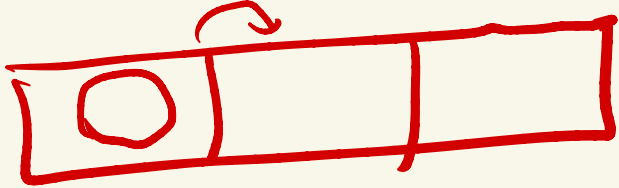
$$n = 3$$



$$\rightarrow 2^1$$



$$\rightarrow 2^2$$



$$\rightarrow 2^3$$

7 8
↑

$$\Rightarrow \sum_{i=1}^n 2^i$$

$$\begin{array}{r} 2^{n+1} - 1 \\ \hline = \sum_{i=1}^n 2^i \end{array}$$

$$2^0 + 2^1 + 2^2 = 2^3 - 1$$

$$x^p = 1$$

$$\sum_{i=0}^n 2^i = \frac{2^{n+1} - 1 - 1}{2 - 2}$$

$$2^{63} = 9 \times 10^{18}$$

$$2^{36} \leq 9 \times 10^{18}$$



Long long

a_1 a_2 a_n n

b_1 b_2 b_n

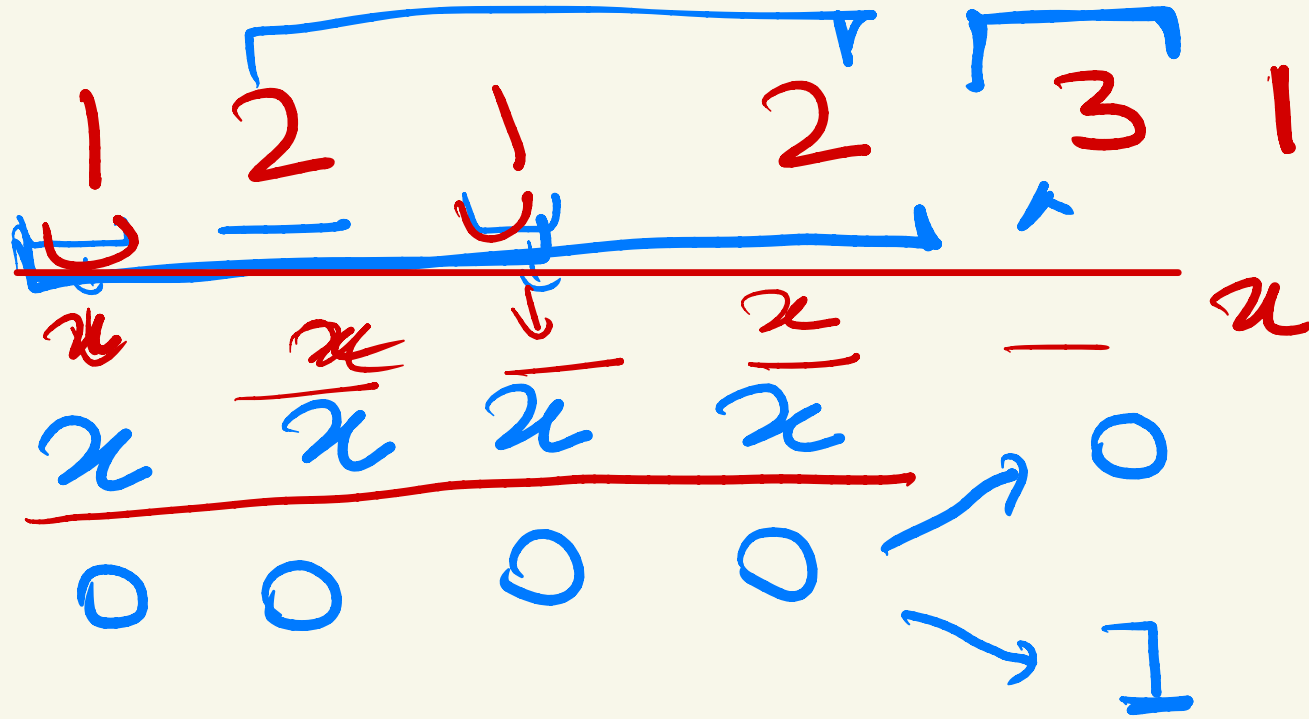
$b_1 = 0$

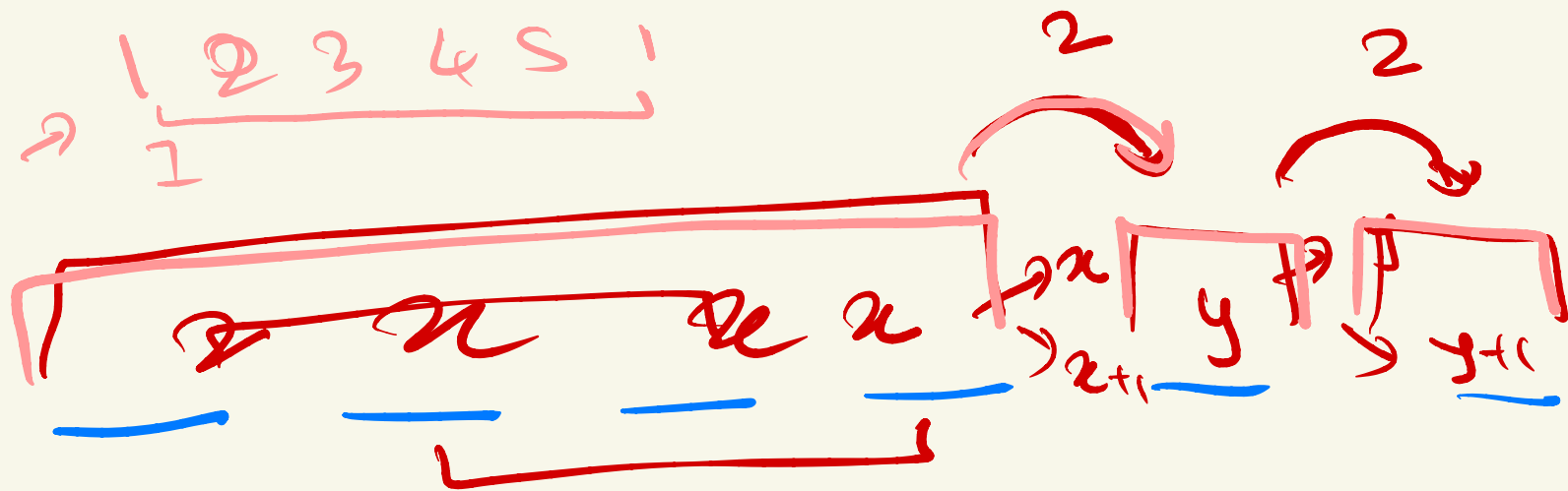
$a_i = a_j \rightarrow (i, j) \quad b_i = b_j$

$b_{i+1} = \underbrace{b_i}_0$

$b_{i+1} = \underbrace{b_{i+1}}_2$

array is sorted.





3

$$2^{3-1} = 2^2$$

ans = 1

ans * = 2

max = 3

2 3 2 3
1 2 1 2
0 1 3 3
3 3

same

3

3

3

3

3

3

1

2

3

1

2

3

1

2

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2

3

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2

3

1

2

3

1

2

3

0

9

dist ≥ 3

0

9

0

0

9

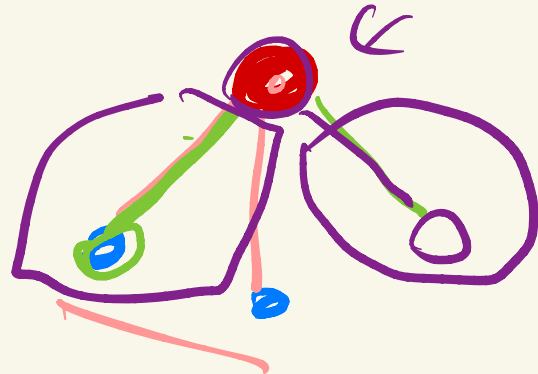
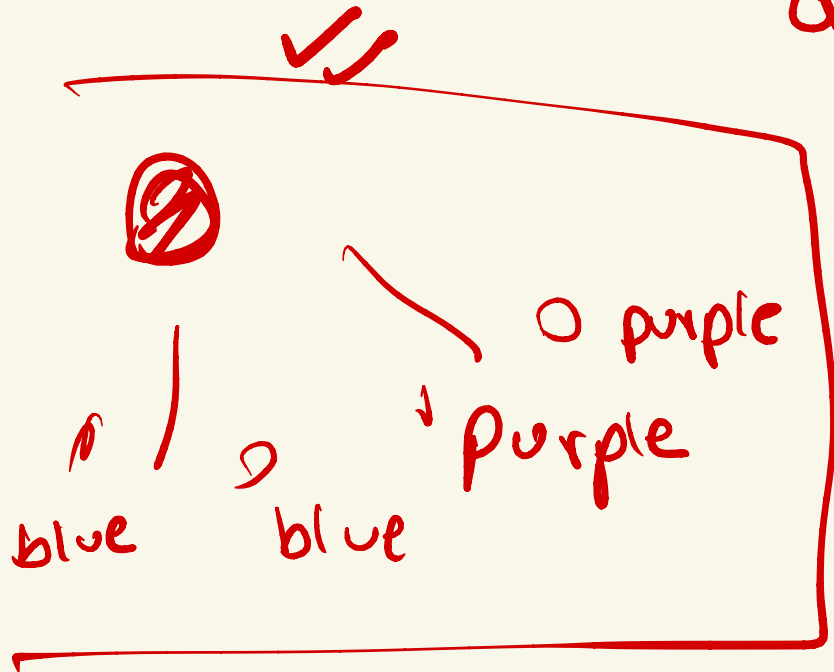
0

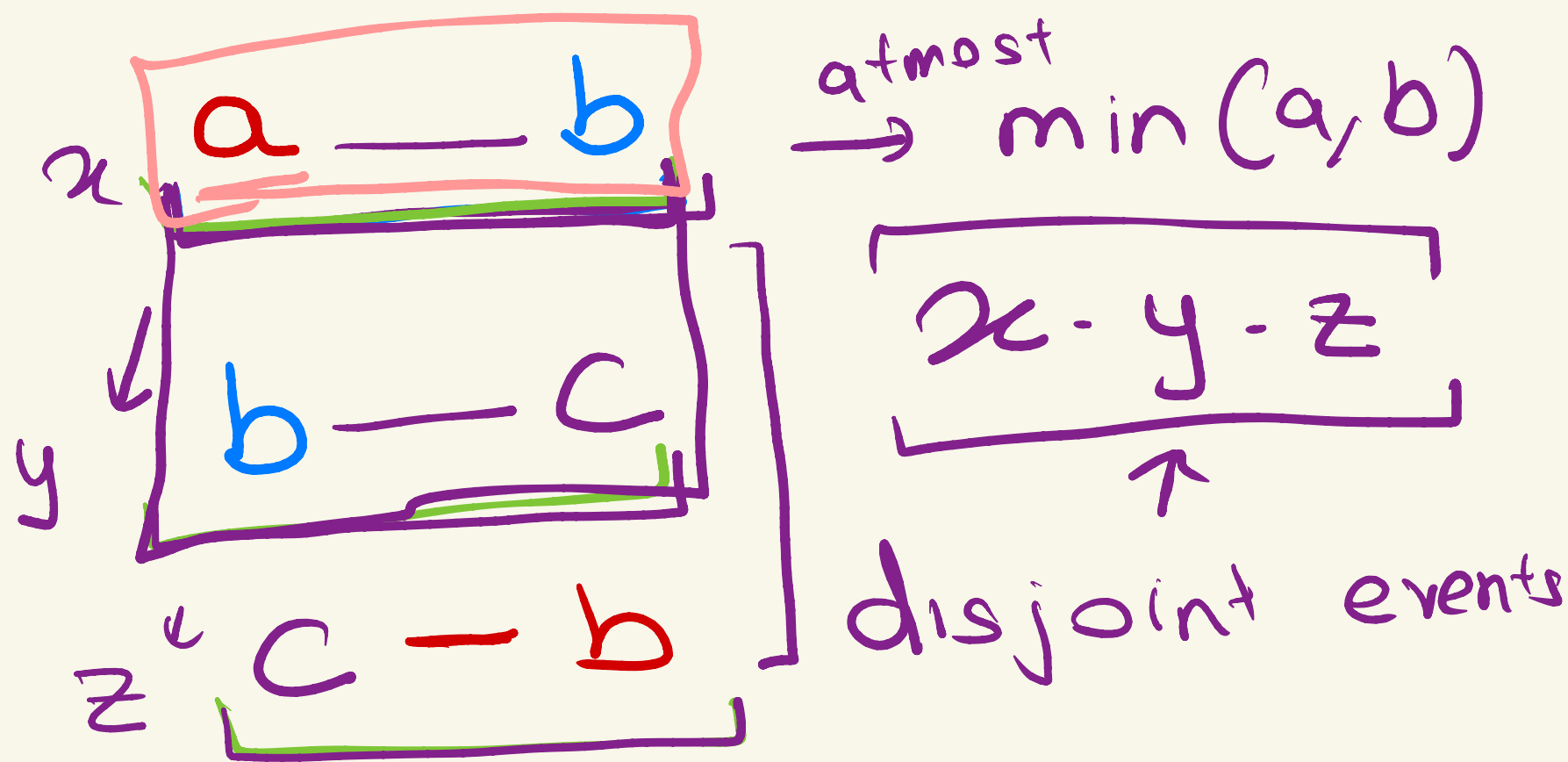
Number of ways to
build bridges

* dist = 1 $\rightarrow$ bridges in same colour

* dist = 2 $\rightarrow$ f r r

a node can be connected to at most 1 node of some color





a

b



$$\sum_{i=0}^{\min(a,b)} \underbrace{a C_i * b C_i}_{T}$$

Bridges

$a C_i \quad b C_i$

$$\begin{array}{c}
 \begin{array}{c} \text{red } n \\ \text{blue } n-1 \end{array} \downarrow \\
 \boxed{[1, 2]} \times \boxed{[1, 2]} \times \boxed{2} \\
 \text{once} \rightarrow \begin{array}{r} 1 - 1 \\ 2 - 2 \end{array} \quad \begin{array}{r} 1 - 2 \\ 2 - 1 \end{array} \\
 \hline
 \boxed{[1, 2][2, 1]}
 \end{array}
 \quad
 \begin{array}{c}
 \text{unord} \\
 \boxed{2} \\
 \hline
 \begin{array}{r} 1 - 2 \\ 2 - 1 \end{array}
 \end{array}
 \quad
 \begin{array}{c}
 \boxed{(1, 2) (2, 1)} \\
 \hline
 \end{array}$$

$$\boxed{[1, 2]}$$

$$\begin{pmatrix} 1 & 2 \end{pmatrix}$$

$$\left[\begin{array}{cc} [1, 2] & [1, 2] \\ [1, 2] & [2, 1] \end{array} \right]$$

$$\begin{array}{r} 1-1 \\ 2-2 \\ \hline \end{array}$$

$$\begin{array}{r} 1-2 \\ 2-1 \\ \hline \end{array}$$

$$[1, 2] [1, 2, 3]$$

$[1, 2, 3]$ $[4, 5, 6]$

$\uparrow \uparrow \uparrow$

$$3 \times 2 \times 1 = \underline{\underline{3!}}$$

${}^a C_3$

$\times$

${}^b C_3$

$$\times \underline{\underline{3!}}$$

More Problems

H.W

- Anagrams [Link](#) ←
- Right Triangles [Link](#) ←
- Count Ways to Make Array With Product [Link](#) ←